Swift Tuition

Uniprocessor Scheduling: Mock Test

Chapter 9 · 30 marks

COS 122 Operating Systems

Swift Tuition | 2026

Before you begin

Write this paper in one sitting before you open the solutions. Total: 30 marks. Suggested time: 45 minutes (**Assumed**: about 1.5 minutes per mark).

Write schedules into the empty boxes or grids: one box (or cell) is one time unit. Show your calculations.

Rules for every question	A process that arrives at time t is ready for the decision at time t. A lower priority number means a higher priority. Waiting time = turnaround time − service time.
Course	COS 122 Operating Systems, Chapter 9: Uniprocessor Scheduling

Questions

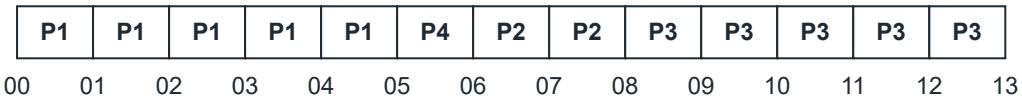
1. Policies from schedules, and round robin	5	p. 3
2. Feedback and HRRN	6	p. 4
3. Shortest remaining time and the measures	7	p. 5
4. Priority scheduling	5	p. 6
5. Types of scheduling	4	p. 7
6. Policy characteristics	3	p. 8
Total	**30**	

Question 1: Policies from schedules, and round robin [5]

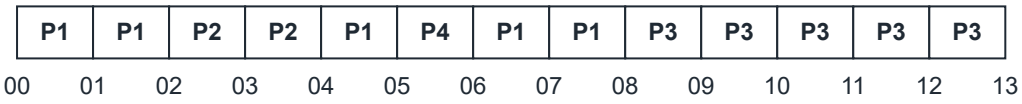
Consider the following workload:

Process	Arrival time	Service time
P1	0	5
P2	2	2
P3	3	5
P4	5	1

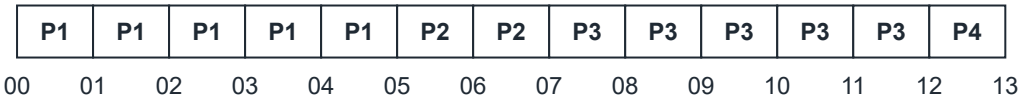
(a) Name a policy that results in the following schedule for the workload given above. **(1)**



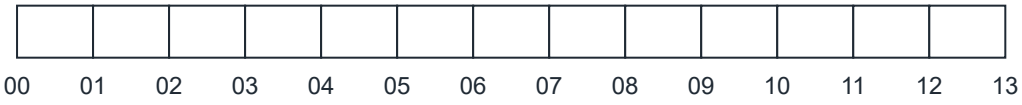
(b) Name a policy that results in the following schedule for the workload given above. **(1)**



(c) Name a policy that results in the following schedule for the workload given above. **(1)**



(d) Show the schedule using the **Round Robin** policy with time quantum **q = 2**. If a quantum ends at the same time as a new process arrives, the new process joins the queue first. Write your solution into the empty boxes of the time scale diagram below. **(2)**



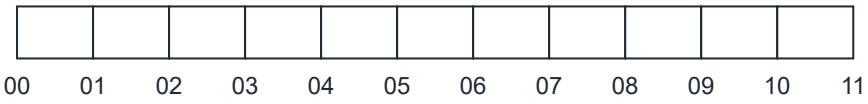
Question 2: Feedback and HRRN [6]

Consider the following workload:

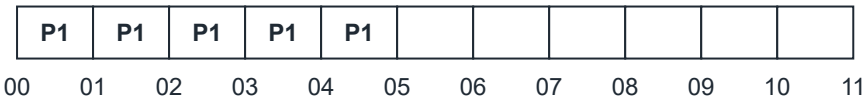
Process	Arrival time	Service time
P1	0	5
P2	1	3
P3	2	2
P4	4	1

(a) Show the schedule using the **Feedback** policy with time quantum $\mathbf{q = 2^i}$. Write your solution into the empty boxes of the time scale diagram below. **(2)**

> **Feedback rules for this question**
>
> - Quantum: $q = 2^i$: RQ0 has q = 1, RQ1 has q = 2, RQ2 has q = 4, RQ3 has q = 8, and so on.
> - A new process joins the back of RQ0 (RQ means ready queue).
> - The scheduler runs the front process of the highest queue that is not empty. Each queue is FCFS.
> - Preemption happens only when a quantum ends. A process that arrives in the middle of a quantum waits.
> - If a process uses its whole quantum and another process is waiting, it moves down one queue. If nothing else is waiting, it stays and starts a new quantum.
> - There is no lowest queue: the number of queues is not limited, so a process that is preempted always moves down one more queue (the textbook rule).



(b) The schedule using the **Highest Response Ratio Next** policy looks as follows after 5 time steps:



(i) Calculate the response ratio of each ready process at time 5. **(3)**

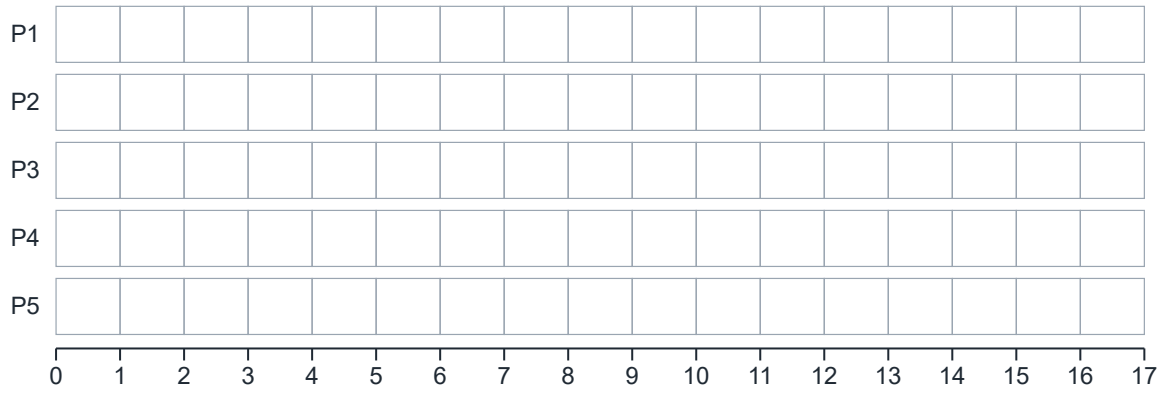
(ii) Which process will be executed next? **(1)**

Question 3: Shortest remaining time and the measures [7]

Consider the following workload:

Process	Arrival time	Service time
P1	0	7
P2	2	4
P3	3	1
P4	5	3
P5	6	2

(a) Draw the schedule using the **Shortest Remaining Time (SRT)** policy on the time-scale grid below. A newly arrived process preempts the running process only if it needs strictly less time than the running process has left. **(3)**



(b) Complete the table for your schedule: finish time, turnaround time T_r, normalised turnaround T_r/T_s and waiting time, and the mean of the last three columns. **(4)**

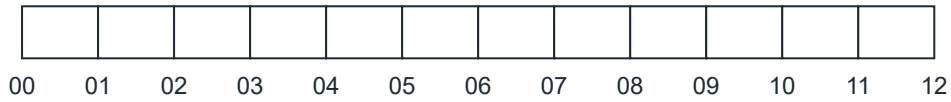
Process	Finish	T_r	T_r/T_s	Waiting
P1				
P2				
P3				
P4				
P5				
Mean				

Question 4: Priority scheduling [5]

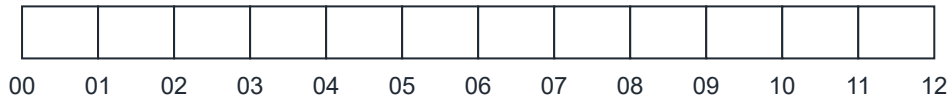
Consider the following workload. A lower priority number means a higher priority.

Process	Arrival time	Service time	Priority
P1	0	4	3
P2	1	3	2
P3	3	2	1
P4	4	3	4

(a) Show the schedule using **non-preemptive priority** scheduling. Write your solution into the empty boxes. **(2)**



(b) Show the schedule using **preemptive priority** scheduling. A new process preempts the running process only if its priority is strictly higher. Write your solution into the empty boxes. **(2)**



(c) Which process in this workload is at risk of starvation under priority scheduling if more high-priority processes keep arriving, and how can an operating system prevent it? **(1)**

Question 5: Types of scheduling [4]

In your own words describe the four types of scheduling. For each type, name the decision it makes. **(4)**

Question 6: Policy characteristics [3]

(a) Give the selection function and the decision mode (non-preemptive, or preemptive and when) of **SRT** and of **HRRN**. Use w for the time spent waiting so far, e for the time spent executing so far and s for the total service time. **(2)**

Policy	Selection function	Decision mode
SRT		
HRRN		

(b) Feedback can use the same quantum q = 1 in every queue, or q = 2^i in queue RQi. Give one difference in how these two versions treat a long process. **(1)**